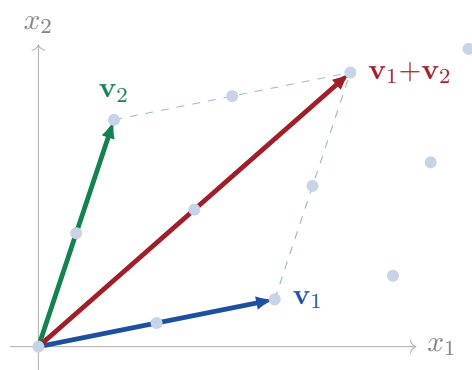


Linear Algebra

Vector Spaces

A Comprehensive Mathematical Treatment

Covering Euclidean spaces, the space $\mathbb{R}^{m \times n}$,
function spaces $C[a, b]$, polynomial spaces P_n ,
the vector space axioms, and fundamental theorems.



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1 Introduction to Vector Spaces

The concept of a *vector space* is one of the most fundamental and unifying ideas in all of mathematics. While the geometric intuition of vectors as “arrows in space” is useful, the formal algebraic structure captures a far richer family of objects — including matrices, continuous functions, and polynomials — under a single coherent framework.

Remark 1.1: Motivation

Before introducing the abstract definition, consider the following objects:

- Ordered n -tuples of real numbers: $\mathbb{R}^n$.
- The set of all $m \times n$ real matrices: $\mathbb{R}^{m \times n}$.
- The set of all continuous functions on $[a, b]$: $C[a, b]$.
- The set of all polynomials of degree less than n : P_n .

Each of these admits natural operations of *addition* and *scalar multiplication* that obey the same algebraic laws. A vector space is precisely the abstract object that encodes those shared laws.

2 Euclidean Vector Spaces

2.1 The Space $\mathbb{R}^n$

Definition 2.1: Euclidean n -Space

The **Euclidean n -space** $\mathbb{R}^n$ is the set of all ordered n -tuples of real numbers:

$$\mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i \}.$$

Equivalently, every element of $\mathbb{R}^n$ may be identified with an $n \times 1$ column vector

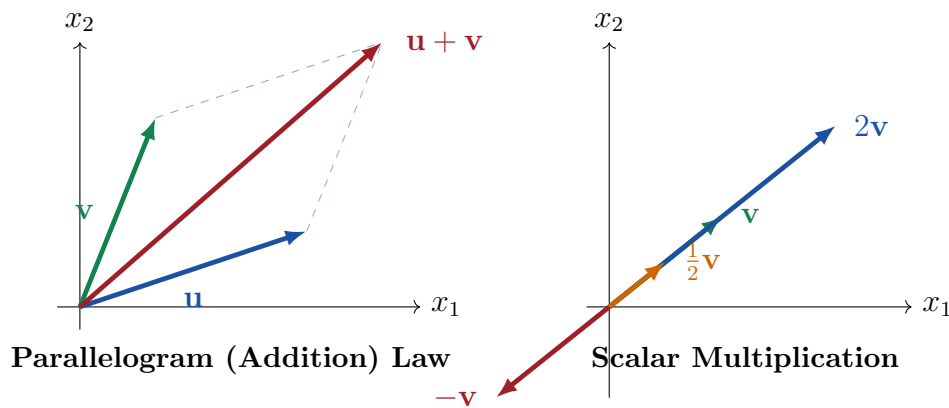
$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Addition and **scalar multiplication** are defined component-wise:

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad \alpha \mathbf{x} = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

2.1.1 Geometric Interpretation in $\mathbb{R}^2$ and $\mathbb{R}^3$

The diagrams below illustrate vector addition (parallelogram law) and scalar multiplication in $\mathbb{R}^2$.



2.2 The Space $\mathbb{R}^{m \times n}$

Definition 2.2: Matrix Space $\mathbb{R}^{m \times n}$

Let $\mathbb{R}^{m \times n}$ denote the set of all $m \times n$ matrices with real entries. For matrices $A = (a_{ij})$ and $B = (b_{ij})$ in $\mathbb{R}^{m \times n}$ and a scalar $\alpha \in \mathbb{R}$, define:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\alpha A)_{ij} = \alpha a_{ij}.$$

The **zero matrix** $\mathbf{0} \in \mathbb{R}^{m \times n}$ is the matrix with all entries equal to zero, and the **additive inverse** of A is $-A = (-a_{ij})$.

To make the definitions concrete, consider two 2×3 matrices:

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & -1 \\ 5 & -3 & 2 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 3 & -2 & 2 \\ 5 & 1 & 1 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & -6 & 9 \\ 0 & 12 & -3 \end{pmatrix}.$$

Remark 2.1: Connection to $\mathbb{R}^n$

The space $\mathbb{R}^n$ is a special case of $\mathbb{R}^{m \times n}$: identifying every $\mathbf{x} \in \mathbb{R}^n$ with its column-vector representation gives the isomorphism $\mathbb{R}^n \cong \mathbb{R}^{n \times 1}$. The operations of addition and scalar multiplication on $\mathbb{R}^n$ are then the matrix operations restricted to $n \times 1$ matrices.

3 The Vector Space Axioms

Having observed that $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ share the same algebraic rules, we now state those rules as *axioms* that define an abstract vector space.

Axioms 3.1: Vector Space Axioms

Let V be a non-empty set equipped with two operations:

- **Addition:** $+: V \times V \rightarrow V$, written $(x, y) \mapsto x + y$.
- **Scalar Multiplication:** $\cdot: \mathbb{R} \times V \rightarrow V$, written $(\alpha, x) \mapsto \alpha x$.

Then V (together with these operations) is called a **vector space over $\mathbb{R}$** if the following eight axioms hold for all $x, y, z \in V$ and all $\alpha, \beta \in \mathbb{R}$:

Label	Name	Statement
A1	Commutativity	$x + y = y + x$
A2	Associativity (add.)	$(x + y) + z = x + (y + z)$
A3	Zero vector	$\exists \mathbf{0} \in V$ such that $x + \mathbf{0} = x$
A4	Additive inverse	$\forall x \in V, \exists -x \in V$ such that $x + (-x) = \mathbf{0}$
A5	Distributivity (scalar)	$\alpha(x + y) = \alpha x + \alpha y$
A6	Distributivity (vector)	$(\alpha + \beta)x = \alpha x + \beta x$
A7	Associativity (scalar)	$(\alpha\beta)x = \alpha(\beta x)$
A8	Identity scalar	$1 \cdot x = x$

The elements of V are called **vectors**; the elements of $\mathbb{R}$ are called **scalars**.

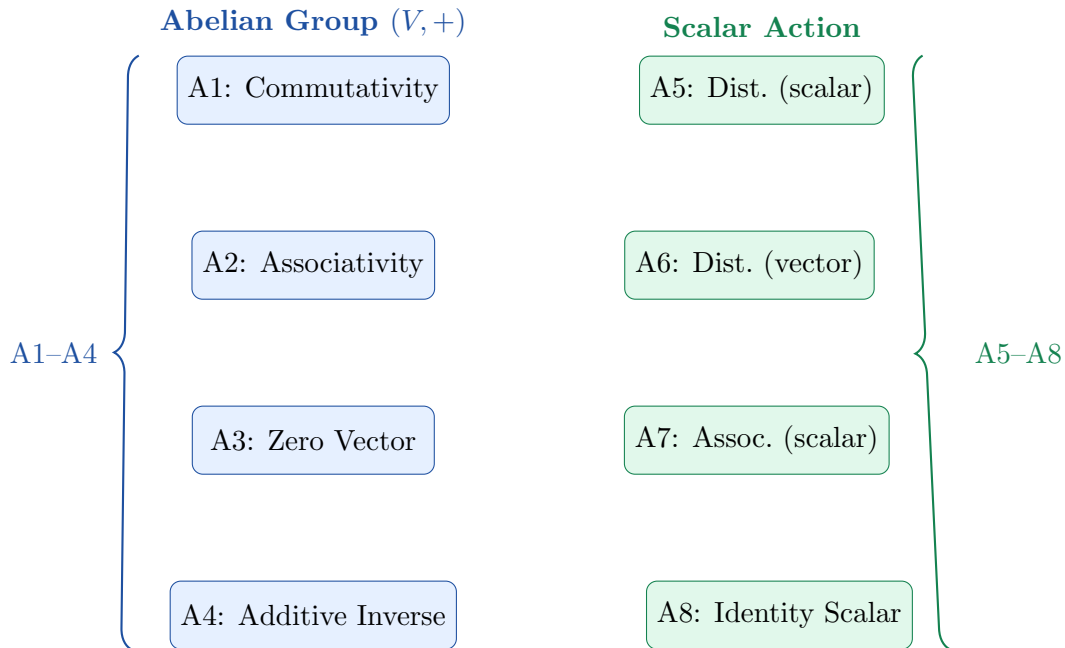
Remark 3.1: Why Eight Axioms?

The eight axioms partition into two groups:

- **Additive axioms (A1–A4):** These say that $(V, +)$ is an *abelian group*.
- **Scalar axioms (A5–A8):** These control how scalars interact with vectors.

Together they capture the essential “linearity” that makes vector spaces so powerful.

The diagram below illustrates the relationships among the axioms:



4 Examples of Vector Spaces

4.1 The Function Space $C[a, b]$

Definition 4.1: $C[a, b]$

Let $C[a, b]$ denote the set of all real-valued functions that are defined and continuous on the closed interval $[a, b]$:

$$C[a, b] = \{ f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous} \}.$$

For $f, g \in C[a, b]$ and $\alpha \in \mathbb{R}$, define

$$(f + g)(x) = f(x) + g(x), \quad (\alpha f)(x) = \alpha f(x), \quad \forall x \in [a, b].$$

Verification of the axioms.

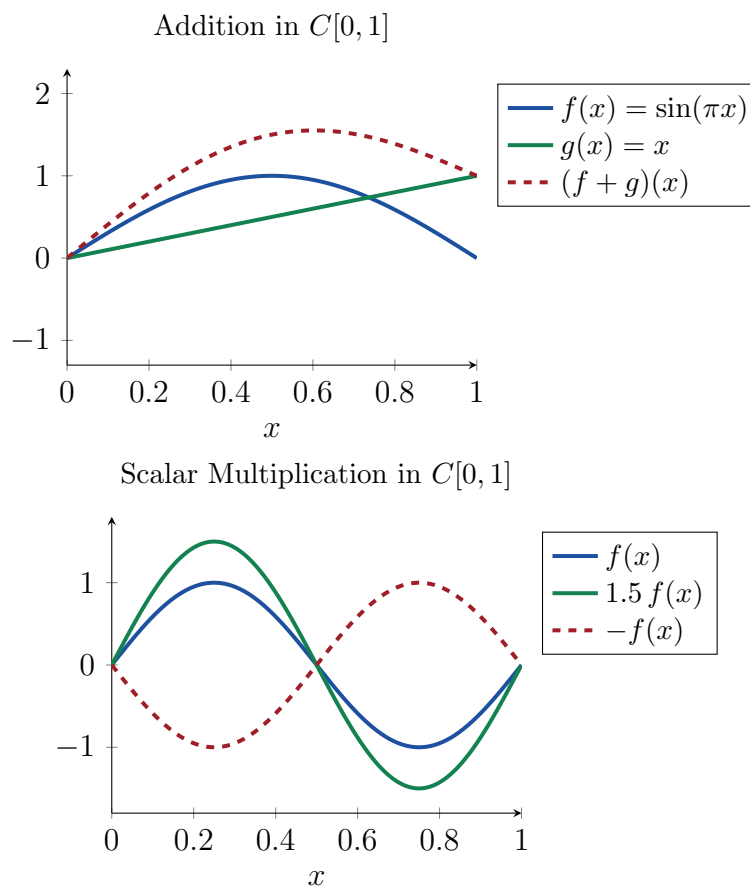
- **Closure under addition:** The sum of two continuous functions is continuous, so $f + g \in C[a, b]$.
- **Closure under scalar multiplication:** αf is continuous since it is a scalar multiple of a continuous function.
- **A1 (Commutativity):** For every $x \in [a, b]$,

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x),$$

so $f + g = g + f$.

- **A3 (Zero vector):** The zero function $z(x) = 0$ for all $x \in [a, b]$ satisfies $(f + z)(x) = f(x) + 0 = f(x)$, hence $f + z = f$.
- **A4 (Additive inverse):** For any $f \in C[a, b]$, the function $(-f)(x) = -f(x)$ is continuous and satisfies $(f + (-f))(x) = f(x) - f(x) = 0 = z(x)$.
- Axioms A2, A5–A8 follow directly from the corresponding properties of real numbers applied pointwise.

The diagram below illustrates addition and scalar multiplication in $C[0, 1]$.



4.2 The Polynomial Space P_n

Definition 4.2: Polynomial Space P_n

Let P_n denote the set of all polynomials of degree *strictly less than* n :

$$P_n = \{ p(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0 \mid a_i \in \mathbb{R} \}.$$

For $p, q \in P_n$ and $\alpha \in \mathbb{R}$, define

$$(p + q)(x) = p(x) + q(x), \quad (\alpha p)(x) = \alpha p(x).$$

The **zero vector** is the zero polynomial

$$z(x) = 0x^{n-1} + 0x^{n-2} + \cdots + 0x + 0.$$

Example 4.1: Addition in P_3

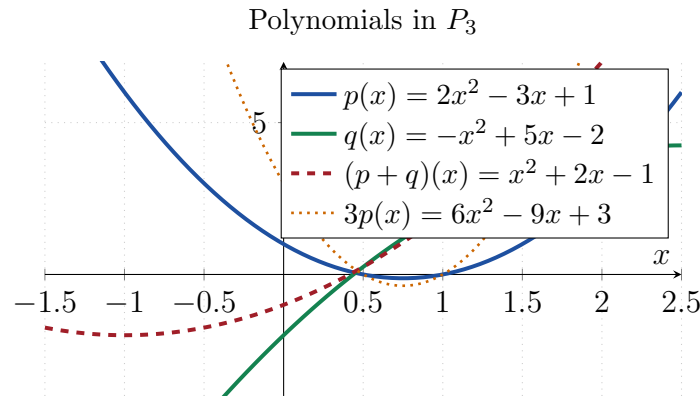
Let $p(x) = 2x^2 - 3x + 1$ and $q(x) = -x^2 + 5x - 2$, both in P_3 . Then:

$$(p + q)(x) = (2x^2 - 3x + 1) + (-x^2 + 5x - 2) = x^2 + 2x - 1,$$

$$(3p)(x) = 3(2x^2 - 3x + 1) = 6x^2 - 9x + 3.$$

Both results lie in P_3 , confirming closure.

The diagram below shows these polynomials graphically.



5 Verification of Axioms: Detailed Walkthrough

To illustrate how the axioms are verified, we walk through the full verification for $\mathbb{R}^{m \times n}$.

Example 5.1: Verifying the axioms for $\mathbb{R}^{m \times n}$

Let $A = (a_{ij}), B = (b_{ij}), C = (c_{ij}) \in \mathbb{R}^{m \times n}$ and $\alpha, \beta \in \mathbb{R}$.

A1. $(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}.$

A2. $((A + B) + C)_{ij} = (a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}) = (A + (B + C))_{ij}.$

A3. The zero matrix $\mathbf{0}$ with all entries $0_{ij} = 0$ satisfies $(A + \mathbf{0})_{ij} = a_{ij} + 0 = a_{ij}.$

A4. $(-A)_{ij} = -a_{ij}$ gives $(A + (-A))_{ij} = a_{ij} - a_{ij} = 0.$

$$\mathbf{A5.} \quad (\alpha(A + B))_{ij} = \alpha(a_{ij} + b_{ij}) = \alpha a_{ij} + \alpha b_{ij} = (\alpha A + \alpha B)_{ij}.$$

$$\mathbf{A6.} \quad ((\alpha + \beta)A)_{ij} = (\alpha + \beta)a_{ij} = \alpha a_{ij} + \beta a_{ij} = (\alpha A + \beta A)_{ij}.$$

$$\mathbf{A7.} \quad ((\alpha\beta)A)_{ij} = (\alpha\beta)a_{ij} = \alpha(\beta a_{ij}) = (\alpha(\beta A))_{ij}.$$

$$\mathbf{A8.} \quad (1 \cdot A)_{ij} = 1 \cdot a_{ij} = a_{ij}.$$

All eight axioms hold, confirming that $\mathbb{R}^{m \times n}$ is a vector space.

6 Fundamental Properties of Vector Spaces

The eight axioms imply several additional useful properties. The theorem below collects the three most basic ones and provides complete proofs.

Theorem 6.1: Additional Properties of Vector Spaces (Theorem 3.1.1)

Let V be a vector space and let $x \in V$. Then:

- (i) $0x = \mathbf{0}$.
- (ii) $x + y = \mathbf{0} \implies y = -x$ (the additive inverse is *unique*).
- (iii) $(-1)x = -x$.

Proof. (i) By axioms **A6** and **A8**:

$$x = 1 \cdot x = (1 + 0) \cdot x = 1 \cdot x + 0 \cdot x = x + 0 \cdot x.$$

Adding $-x$ to both sides:

$$\mathbf{0} = -x + x = -x + (x + 0 \cdot x) \stackrel{\mathbf{A2}}{=} (-x + x) + 0 \cdot x \stackrel{\mathbf{A4}}{=} \mathbf{0} + 0 \cdot x \stackrel{\mathbf{A3}}{=} 0 \cdot x.$$

(ii) Suppose $x + y = \mathbf{0}$. Then:

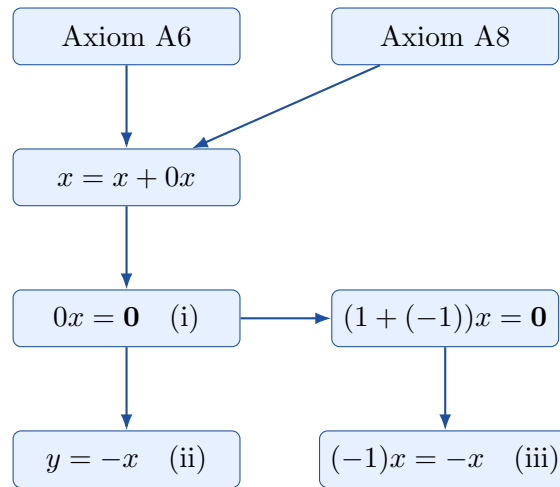
$$\begin{aligned} -x &= -x + \mathbf{0} \stackrel{\mathbf{A3}}{=} -x + (x + y) \\ &\stackrel{\mathbf{A2}}{=} (-x + x) + y \stackrel{\mathbf{A4}}{=} \mathbf{0} + y \stackrel{\mathbf{A1}, \mathbf{A3}}{=} y. \end{aligned}$$

(iii) By part (i) and axiom **A6**:

$$\mathbf{0} = 0 \cdot x = (1 + (-1)) \cdot x \stackrel{\mathbf{A6}}{=} 1 \cdot x + (-1) \cdot x \stackrel{\mathbf{A8}}{=} x + (-1) \cdot x.$$

Thus $x + (-1) \cdot x = \mathbf{0}$, and by part (ii), $(-1) \cdot x = -x$. ■

The logical flow of the proof is summarised in the following diagram:



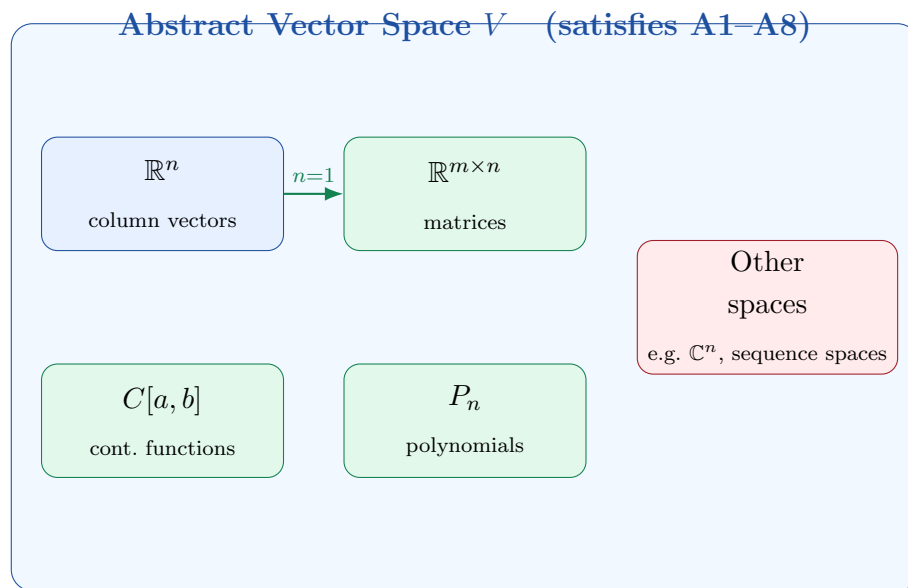
7 Comparison of Vector Space Instances

The following table summarises the key attributes of each vector space discussed in this document.

Space	Elements (vectors)	Zero vector	Dimension
$\mathbb{R}^n$	n -tuples $(x_1, \dots, x_n)$	$(0, \dots, 0)$	n
$\mathbb{R}^{m \times n}$	$m \times n$ matrices	$m \times n$ zero matrix	mn
$C[a, b]$	Continuous functions on $[a, b]$	$z(x) = 0$	∞
P_n	Polynomials of degree $< n$	$z(x) = 0$	n

8 Illustrative Summary Diagram

The diagram below places the four vector spaces within the broader landscape of abstract vector spaces.



9 Further Consequences of the Axioms

Beyond Theorem 6.1, the axioms yield the following important results. Their proofs are left as guided exercises.

Theorem 9.1: Additional Consequences

Let V be a vector space, $\alpha \in \mathbb{R}$, and $x \in V$.

- (i) $\alpha \mathbf{0} = \mathbf{0}$.
- (ii) $\alpha x = \mathbf{0} \implies \alpha = 0$ or $x = \mathbf{0}$.
- (iii) The zero vector $\mathbf{0}$ is unique.

Proof of (i). By axiom A3, $\mathbf{0} + \mathbf{0} = \mathbf{0}$. Multiplying by α and applying A5:

$$\alpha \mathbf{0} = \alpha(\mathbf{0} + \mathbf{0}) = \alpha \mathbf{0} + \alpha \mathbf{0}.$$

Adding $-(\alpha \mathbf{0})$ to both sides (using A4 and A3):

$$\mathbf{0} = \alpha \mathbf{0}.$$

■

Proof of (iii) (Uniqueness of the zero vector). Suppose $\mathbf{0}_1$ and $\mathbf{0}_2$ both satisfy A3. Then:

$$\mathbf{0}_1 = \mathbf{0}_1 + \mathbf{0}_2 = \mathbf{0}_2 + \mathbf{0}_1 = \mathbf{0}_2,$$

where the first and third equalities use A3 for $\mathbf{0}_2$ and $\mathbf{0}_1$ respectively, and the middle equality uses A1. ■

10 Summary

The key ideas of this chapter may be summarised as follows:

Chapter Summary

- A **vector space** over $\mathbb{R}$ is a set V with addition and scalar multiplication satisfying axioms A1–A8.
- The Euclidean space $\mathbb{R}^n$, the matrix space $\mathbb{R}^{m \times n}$, the function space $C[a, b]$, and the polynomial space P_n are all **concrete examples** of vector spaces.
- The **zero vector** is unique; every element has a **unique additive inverse**; and $(-1)x = -x$.
- The scalar 0 always maps any vector to $\mathbf{0}$: $0 \cdot x = \mathbf{0}$.
- The abstract framework unifies seemingly different mathematical objects under a single set of algebraic rules, enabling results proved once to apply universally.